



Degradation modeling under time-varying operating conditions: Inference and prognosis with particle filter

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ABSTRACT

The health state estimation and remaining useful life prediction have always been challenging when a mechanical system works under time-varying operating conditions (TVOC). TVOC can not only influence the amplitude of degradation growth, but also have an impact on how the baseline degradation process is observed. In this paper, we propose a state-space model to estimate the health state of a single-unit system under continuous TVOC. Compared to most existing works where only one discrete operating condition is considered, our model accounts for multi-dimensional and continuous TVOC. The baseline (hidden) degradation is described by stochastic processes or general path models, and the observations are subject to TVOC-related observation bias and noise. Given the raw observations and the TVOC sequences, a particle filter (PF) is used to estimate the hidden degradation process. When the model parameters are unknown, Monte Carlo Expectation-Maximization is used for parameter estimation. As a result, the health state at any time and any TVOC can be estimated. The proposed method is applied to a real choke valve erosion dataset, showing a significant improvement in the health state estimation compared to traditional methods.

1. Introduction

Mechanical systems are sensitive and prone to degradation resulting from the direct interaction with dynamic working conditions. How to estimate the system state attracts increasing attention, since their failures have the potential for inflicting equipment damages, economic loss, and even catastrophic disasters. The advancement of sensing technology boosts the evolution from failure data-based to more dynamic condition-monitoring data-driven management. The latter has the advantage of enabling more accurate remaining useful lifetime (RUL) prevention by leveraging the diagnostic and prognostic of the collected system condition data, which is essential for production and maintenance managers. Nevertheless, many existing RUL prediction methods are based on the assumption of a stable operating condition or its limited impact on the degradation process [1–3]. These over-simplified assumptions are barely met in reality.

An example is the subsea choke valves. In O&G industry, production chokes are designed to take the brunt of the pressure off the line components, increasing their life and yielding significant benefits. By

restricting the flow to a small opening or orifice, a choke valve reduces the well pressure, controls production rate, creating downstream or back pressure. They stand out as the components in oil & gas production systems that are most susceptible to erosion [4], due to the potential for high flow velocities created by the pressure let-down across the choke, and the exposure to the abrasive well stream that consists of a mix between oil, gas, water, and sand.

One health indicator reflecting the erosion status of the choke valves is the difference between the theoretical and estimated valve flow coefficients (Cv) [4], which is referred to as Cv deviation in the rest of the paper. For certain valves, e.g., multiple-orifice valve (MOV) or multistage/labyrinth cage with interval plug choke, Cv deviation can reveal the level of damage since the labyrinths wear gradually, manifesting its suitability as a suitable indicator in condition monitoring. From a practical perspective, the choke valve may operate at a different opening each day. It implies that the Cv deviation not only changes with time (due to erosion, corrosion, and clogging) but also depends on

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the valve opening (related to the practical working condition). In fact, there is a Cv curve (as a function of valve opening) that gradually drifts away from the theoretical Cv curve each day. The erosion ground truth is therefore a surface that can never be observed, since it is unrealistic to measure the Cv deviation at different openings in a single day. Therefore, traditional degradation path models or stochastic process models such as Gamma process [5,6] or its extensions [7] may not be suitable to describe the evolution of the Cv deviation, unless the valve opening is restricted to a small range during the observation period. Note that, although the wear and erosion of a choke valve have also been investigated from a material [8–10] or flow [11] perspective, these methods mainly tend to guide the design of choke valves which limits their applicability for online condition monitoring of the choke valves in the operational phase.

In the framework of condition-based maintenance and RUL estimation, data-driven methods are becoming increasingly popular due to their capabilities in learning the degradation characteristics from training data of failed units. For example, the RUL prediction for slewing bearing in different working conditions has been investigated in [12], based on the similarity between service sample and reference sample. Recurrent neural network, convolutional neural network and LSTM has been employed to predict the RUL of rolling bearings in [13–15]. Most recently, authors in [16] proposed a novel variational auto-encoder-long short-term memory network-local weighted deep sub-domain adaptation network (VLSTM-LWSAN) for RUL prediction to overcome existing limitations in domain adaptation learning-oriented transfer learning, such as offline data and online data must obey a similar distribution. Typically, a large amount of failure event data is needed for data-driven approaches. In practice, however, it is hard to gather those data as many systems are not allowed to run to failure, e.g., choke valves or wind turbines [17]. Furthermore, the degradation processes of different choke valves exhibit heterogeneity, due to their unique operation profiles and external environments. It is more reasonable to predict the degradation trend of a choke valve according to its own historical degradation trajectory.

We present in this paper a novel model-based prognosis method for systems subject to TVOC. In the following, the existing literature on this topic is reviewed before highlighting our contributions.

1.1. Literature review

Generally, the effects of TVOC on mechanical systems RUL estimation can be classified into two categories: (1) degradation rate changes with different operating conditions; and (2) degradation level jumps at condition changing points. In addition, TVOC can have an impact on the lifetime distribution as in [18–20], but that is not the primary focus of this paper. A framework for the RUL prediction of the single-unit system, assuming with a Brownian motion evolving in response to the TVOC, were proposed by the combination of in situ degradation measurements and operating conditions [21]; the proposed model was further developed to predict the RUL in both the piecewise constant and stochastic working condition. Authors in [22] integrated three sources of information, including field degradation measurements, real-time operating conditions and accelerated degradation testing (ADT) data, to state space modeling based on Wiener process with nonlinearity. An acceleration factor was introduced to link the operating condition and degradation rate. The joint dependency of degradation rate and TVOC was quantified and considered with the characterization of the degradation process of the component into a state function and an observation function [23]. Authors in [24] proposed a Wiener process-based model to account for the dependency of the degradation process on the system's operation states, in which a two-state homogeneous continuous-time Markov chain (CTMC) was used to approximate the transition process between the system operating state and the storage state. Similarly, TVOC is modeled as a homogeneous CTMC in [25].

With the integration of historical and real-time signals and an underlying degradation process, TVOC were thus considered in the estimation of system residual life distributions. A RUL estimation method for periodically monitored system was proposed in [26] and the system's degradation rate is influenced by specific operational condition. Moreover, TVOC were technologically evolved as a discrete-time Markov chain (DTMC). A semi-Markov decision process framework was proposed in [27] to estimate the average RUL of continuously degraded systems under dynamic operating conditions that evolves under DTMC. Most recently, authors in [28] proposed a stochastic hybrid system model, where the operating condition is modeled as a CTMC, to predict the remaining useful life. Similar works also exist in ADT and vibration analysis [29–31]. In most of the aforementioned literature, the TVOC is discrete and does not cause degradation level shifts.

The second research line receives less attentions. Authors in [32] jointly considered both TVOC-dependent degradation rate and signal jumps at changing points. Under a state-space modeling framework, a measurement function is used to describe the observed degradation paths and delivered closed-form solutions for the distribution of RUL. The operating conditions were assumed discrete, and a signal transformation algorithm was proposed to transform the degradation signals under different operating conditions into a baseline condition.

To overcome the limitation that TVOC is discrete, authors in [33] investigated the RUL prediction for wind turbines under continuously varying operating condition, i.e., rotating speed. The raw observations measured at different timestamps under different speed are first converted to a baseline condition using a transformation function, then the parameters of the baseline degradation path (which is now considered fully observed) are estimated. RUL is also derived based on the baseline degradation.

However, the method in [33] still bears some drawbacks and limitations. First, selecting a baseline operating condition can sometimes be difficult when there is not a “baseline” operating condition under which the system works most often. Second, their model is specifically built for wind turbines in the sense that the degradation increment depends on the *accumulation* of an operating condition rather than the instantaneous value. This modeling may not be suitable for other systems. Third, the parameter estimation relies on presumed reference values, which is not credible if there are too few measurements taken under the baseline condition or if those measurements are contaminated by noise.

1.2. Contributions

In this paper, a state-estimation method for systems working under continuous TVOC is presented. We model the TVOC as continuous variables. The state-space model explicitly describes the relationship between the observed HI and TVOC. More precisely, in the state-transition model, the distribution of the baseline degradation increment is assumed to depend on TVOC; in the observation model, TVOC is assumed to influence both the initial values of the HI and the measurement noise. A particle filter (PF, [34]) is used to estimate the hidden degradation. The model is inspired by a phenomenon that operating subsea choke valves at smaller openings will often lead to more severe Cv deviations and larger measurement errors. And the valves are generally not good-as-new at the beginning of their service, so the initial Cv deviation at different openings is different. A parameter estimation method based on the PF and expectation-maximization (EM) is also proposed.

The unique features of this paper compared to existing works [32, 33] are stated below. First, we extend the discrete operating conditions in [32] to the continuous case, and assume additionally that TVOC may influence the distribution of measurement noise, which makes the model more realistic. This leads to a fundamental difference in the parameter estimation procedure: discrete TVOC often allows the parameters in the state transition model and observation model to be

estimated separately, whereas continuous TVOC implies the use of PF-based EM, because the parameters in the state transition model and observation model are intertwined. Second, compared to the method in [33] which is more tailored to RUL prediction of wind turbines, the model presented in this paper extends their approach by including (1) multiple TVOC and (2) a general function f that is used to describe the influence of TVOC on the transition density (degradation rate). Given that [33] assumes that the integration of rotation speed of wind turbines is the only factor that influences the degradation rate, our model is more general and flexible. And finally, our model does not rely on the transformation of the raw observation to the baseline condition, so the parameter estimation can be more robust facing measurement noise and data scarcity.

1.3. Scope and limitations

This research aims to infer the degradation “masked” by the time-varying operating conditions. The primary focus is placed on model building and parameter estimation. An example of RUL estimation is presented in the case study. However, since RUL’s distribution depends heavily on TVOC, which is supposed to determine the degradation rate, a thorough discussion of RUL estimation based on modeling the future evolution of environmental/controllable operating conditions is valuable, but is not included in this work. This research also assumes that field experts are fully capable of identifying the most appropriate model structures. Statistic-based goodness-of-fit analysis for model comparison and validation as in [33] is not considered here. A flowchart of the proposed method is illustrated in Fig. 1, outlining the input, output, and key steps of our model.

The rest of the paper is organized as follows: the model formulation is detailed in Section 2, with an example of using PF to estimate the hidden Gamma baseline degradation. Parameter estimation using Monte Carlo expectation maximization (MCEM) is presented in Section 3. Section 4 presents a case study on the real industrial choke valve erosion data, and the whole procedure of parameter and degradation estimation is elaborated. The strength of the proposed method has been demonstrated through the comparison with Wiener process, noisy Gamma process and the method in [33]. Discussions on the health monitoring for choke valves are also provided. Finally, concluding remarks are given in Section 6.

2. Model formulation

Let p be the total number of operating conditions under consideration and let $\mathbf{v}$ be a vector of length p , where $\mathbf{v}[j], j = 1 \dots p$, is the j th operating condition. The vector $\mathbf{v}$ can contain one single element or multiple elements, e.g., the rotating speed of wind turbines [33] or the opening of choke valves. A scale transformation is made so that at any instant t , the $\mathbf{v}[j]$ takes a value in the closed interval $[0, 1]$, with 0 and 1 representing the possible minimal and maximal value of $\mathbf{v}[j]$. The scaling procedure aims to avoid numerical errors or overflows in PF smoothing/EM inference. At time 0, the initial value of a health indicator is a function defined on $[0, 1]^p$ which we denote as $\phi(\mathbf{v}|\theta_\phi)$, with θ_ϕ the associated parameter vector. In the simplest cases, ϕ can be a straight line or a plane. If there is no TVOC, then $p = 0$, and ϕ reduces to a point.

Let X_t be the baseline degradation process, e.g., a Gamma or Wiener process, with $X_0 = 0$. X_t describes how the health indicator evolves at the baseline operating condition $\mathbf{0}$. We assume a translation effect of X_t on $\phi(\mathbf{v})$, that each point in the operating condition space is subject to the same degradation: none degrades slower or faster than others. This is similar to the location transformation function in [33]. Notations and abbreviations used in this paper are presented in Table 1.

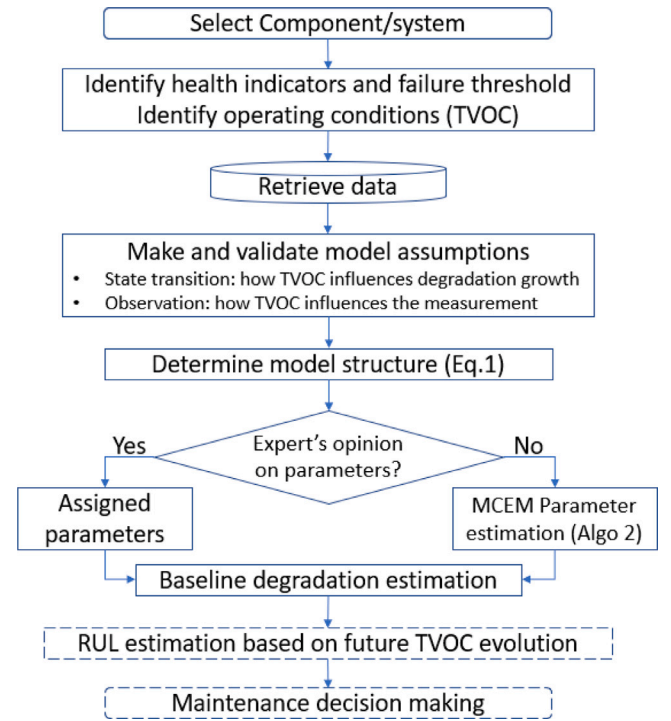


Fig. 1. Flowchart of the proposed method.

Table 1

Notations and abbreviations.

Notation	Interpretation
TVOC	Time-varying operating conditions
PDF	Probability density function
RUL	Remaining useful life
PF	Particle Filter
HI	Health indicator
Cv	Flow coefficient of a choke valve
MCEM	Monte Carlo expectation-maximization
p	Number of TVOC
$\mathbf{v}$	TVOC values
X, x	Baseline degradation process
Y, y	Observed degradation process
z	Baseline degradation paths sampled from a PF
$\theta = [\theta_x, \theta_\phi, \sigma]$	Parameter vector
ϵ	Measurement error
ϕ	Initial degradation level
$\Delta\phi_k$	Observation bias due to the change in the operating condition
σ	Standard deviation of the measurement error
f	Distribution of baseline degradation increment
$f(x_t x_{t-1})$	Transition density
$g(y_t x_t)$	Observation density
π	Joint distribution
q	Instrumental distribution
$w, \tilde{w}$	Normalized and unnormalized particle weight
α, β, η	Parameters in a Gamma process
Γ	Gamma function
τ	Rank correlation: Kendall's τ
D	Failure threshold
L	Log-likelihood function
N	Number of particles used in a PF
M	Number of iterations in MCEM
n	Number of observations
U, V	Column vector and matrix formed by $\mathbf{v}$, used in MCEM
W	Brownian motion
Λ	Time transformation in a Wiener process

2.1. State-space model representation

Suppose that the HI is monitored at discrete timestamps $t_{0:n} = \{t_0, t_1, \dots, t_n\}$. Let $\mathbf{v}_{0:n} = \{\mathbf{v}_0, \mathbf{v}_1, \dots, \mathbf{v}_n\}$ be the operating conditions at time $t_0, t_1, \dots, t_n$. Let $\mathbf{x}_{0:n} = \{x_0, x_1, \dots, x_n\}$ be the unobserved baseline degradation process and $\mathbf{y}_{0:n} = \{y_0, y_1, \dots, y_n\}$ be the corresponding observed HI. The following state-space model can be developed:

$$\begin{cases} x_k = x_{k-1} + \Delta x_k \\ y_k = x_k + \Delta \phi_k + \epsilon_k \end{cases} \quad (1)$$

where Δx_k is the k th baseline degradation increment. Its distribution is determined by the parameter vector θ and the TVOC on the previous moment, $\mathbf{v}_{k-1}$:

$$\Delta x_k \sim f(x|\theta, \mathbf{v}_{k-1}) \quad (2)$$

For example, for choke valves, f could be a normal distribution whose mean is a function of the valve opening.

$\Delta \phi_k = \phi(\mathbf{v}_k|\theta) - \phi(\mathbf{v}_{k-1}|\theta)$ is the observation bias due to the change in the operating condition. ϵ_k is a Gaussian distributed noise with mean 0 and TVOC-dependent standard deviation: $\epsilon_k \sim \mathcal{N}(0, \sigma(\mathbf{v}_k|\theta))$.

The system is fully determined by the functions ϕ , f , σ and the parameter θ . When the parameters are known in advance, the hidden state $\mathbf{x}_{0:n}$ can be estimated from the noisy observations using a PF.

2.2. State estimation using particle filter

Given the observations $\mathbf{y}_{0:t}$, our interest lies in finding or sampling from the posterior distribution $\pi_{0:t|0:t}(\mathbf{x}_{0:t}|\mathbf{y}_{0:t})$. This can be achieved by the prediction–correction recursion:

$$\pi_{0:t|0:t-1}(\mathbf{x}_{0:t}|\mathbf{y}_{0:t-1}) = \pi_{0:t-1|0:t-1}(\mathbf{x}_{0:t-1}|\mathbf{y}_{0:t-1})f(x_t|\mathbf{x}_{t-1}) \quad (3)$$

and

$$\pi_{0:t|0:t}(\mathbf{x}_{0:t}|\mathbf{y}_{0:t}) = \frac{g(y_t|\mathbf{x}_t)\pi_{0:t|0:t-1}(\mathbf{x}_{0:t}|\mathbf{y}_{0:t-1})}{\pi_{0:t-1|0:t-1}(\mathbf{y}_t|\mathbf{y}_{0:t-1})} \quad (4)$$

where $\pi_{0:t-1|0:t-1}(\mathbf{y}_t|\mathbf{y}_{0:t-1})$ is a normalizing constant (independent of the state) for a fixed data realization. Sampling directly from $\pi_{0:t|0:t}(\mathbf{x}_{0:t}|\mathbf{y}_{0:t})$ can be difficult, thus, a sequential version of the importance sampling and resampling procedure outlined in Eq. (3) and (4) are carried out, which forms the basis of a PF. We recall in the following only the key steps in a PF. Readers may refer to [35] for more details.

The idea is to generate N particle paths $\mathbf{x}_{0:t}^{(i)}, i = 1 \dots N$ from an instrumental (or importance) distribution $q(\mathbf{x}_{0:t}|\mathbf{y}_{0:t})$ and compute the unnormalized importance weights:

$$\tilde{w}_t^{(i)} = \frac{\pi_{0:t|0:t}(\mathbf{x}_{0:t}^{(i)}|\mathbf{y}_{0:t})}{q_{0:t}(\mathbf{x}_{0:t}^{(i)}|\mathbf{y}_{0:t})}$$

which can be normalized to obtain $w_t^{(i)} = \tilde{w}_t^{(i)} / \sum_{j=1}^N \tilde{w}_t^{(j)}$. Note that, the instrumental distribution can be constructed in a way that allows sequential computations:

$$q_{0:t}(\mathbf{x}_{0:t}|\mathbf{y}_{0:t}) = q_{0:t-1}(\mathbf{x}_{0:t-1}|\mathbf{y}_{0:t-1})q_t(x_t|\mathbf{x}_{t-1}, \mathbf{y}_t)$$

where $q_t(x_t|\mathbf{x}_{t-1}, \mathbf{y}_t)$ can be regarded as an extension of the existing path. Thus, in a PF, the following two steps are iteratively conducted:

- Propagate particles: sampling $\mathbf{x}_t^{(i)}$ from $q_t(\mathbf{x}_t|\mathbf{x}_{t-1}^{(i)}, \mathbf{y}_t)$
- Compute weight: $\tilde{w}_t^{(i)} = w_{t-1}^{(i)} \frac{g(y_t|\mathbf{x}_t^{(i)})f(\mathbf{x}_t^{(i)}|\mathbf{x}_{t-1}^{(i)})}{q_t(\mathbf{x}_t^{(i)}|\mathbf{x}_{t-1}^{(i)}, \mathbf{y}_t)}$

The sequential importance sampling approach described above bears the drawback of particle degeneration: a few particles contain nearly all of the probability mass, and hence most particles have 0 weight. Thus, resampling should be applied in every step to eliminate those particles with a small weight. The basic method consists in sampling N draws from the current population using the normalized weight $w_t^{(i)}$ for each i . In the rest of the paper, we use Algorithm 3 in [35] to implement PF.

2.3. A gamma baseline degradation process as an example

In this section, we present an artificial example where the increments in the baseline degradation are governed by a Gamma distribution. The k th increment is distributed as

$$f(x|\theta, \mathbf{v}_{k-1}) = \frac{\beta^{\alpha_k}}{\Gamma(\alpha_k)} x^{\alpha_k-1} \exp(-\beta x) \quad (5)$$

where α_k and β are respectively the shape and rate parameter of the Gamma distribution. α_k is determined by a baseline shape parameter α , the time increment $t_k - t_{k-1}$, a proportional factor η and the most recent operating condition $\mathbf{v}_{k-1}$:

$$\alpha_k = \alpha(t_k - t_{k-1}) \exp(-\eta \|\mathbf{v}_{k-1}\|_2) \quad (6)$$

where $\|\cdot\|_2$ represents the L^2 norm.

Assume the following functional forms for the observation model:

$$\begin{cases} \phi(\mathbf{v}_k|\theta) = \mathbf{v}_k^T \theta \\ \sigma_k = \sigma \|\mathbf{v}_k\|_2 \end{cases} \quad (7)$$

so that ϕ is a linear function of the TVOC and that the measurement error is proportional to the L^2 norm of TVOC. The parameter vector is therefore $\theta = [\alpha, \beta, \eta, \theta_\phi, \sigma]$. The observation density $g(y_k|\mathbf{x}_k)$ is a normal distribution:

$$g(y_k|\mathbf{x}_k) = \frac{1}{\sigma \|\mathbf{v}_k\|_2 \sqrt{2\pi}} \exp\left(-\frac{(y_k - x_k - \Delta \phi_k)^2}{2(\sigma \|\mathbf{v}_k\|_2)^2}\right) \quad (8)$$

We use the *optimal kernel* [36] as the proposal distribution, which can be recognized as the conditional distribution of the hidden state $\mathbf{x}_k$ given $\mathbf{x}_{k-1}$ and the current observation $\mathbf{y}_k$:

$$\begin{aligned} q_k(\mathbf{x}_k|\mathbf{x}_{k-1}, \mathbf{y}_k) &= \frac{f(\mathbf{x}_k|\mathbf{x}_{k-1})g(\mathbf{y}_k|\mathbf{x}_k)}{\int f(\mathbf{x}|\mathbf{x}_{k-1})g(\mathbf{y}_k|\mathbf{x})d\mathbf{x}} \\ &\propto (x_k - x_{k-1})^{\alpha_k-1} \exp(-\beta(x_k - x_{k-1})) \exp\left(-\frac{(y_k - x_k - \Delta \phi_k)^2}{2(\sigma \|\mathbf{v}_k\|_2)^2}\right) \end{aligned} \quad (9)$$

Sampling from Eq. (9) requires techniques such as rejection sampling. We introduce first the generalized Gamma distribution as in Eq. (10). Then $\mathbf{x}_k$ can be sampled according to Algorithm 1. This should be faster as the generalized gamma distribution is implemented in most programming tools.

$$Gamma(x, a, c, u) = u^a (x - c)^{a-1} e^{-u(x-c)}. \quad (10)$$

Algorithm 1 Rejection sampling

- 1: Generate $Z \sim Gamma(x, \alpha_k, x_{k-1}, \beta)$
- 2: Generate $U \sim U[0, 1]$
- 3: if $U \leq \exp\left(-\frac{(y_k - Z - \Delta \phi_k)^2}{2(\sigma \|\mathbf{v}_k\|_2)^2}\right)$, set $X_k = Z$. Otherwise, return to 1.

The weight of the i th particle is computed by Eq. (11), which will then be used for resampling.

$$\tilde{w}_k = \tilde{w}_{k-1} \int f(\mathbf{x}|\mathbf{x}_{k-1})g(\mathbf{y}_k|\mathbf{x})d\mathbf{x} \quad (11)$$

3. Parameter estimation

It is unfeasible to estimate the model parameters using direct MLE with the latent variables, which here refer to the unobserved baseline degradation process, measurement errors, and the observation bias related to TVOC. In this case, the expectation–maximization (EM) [37] and its variations such as stochastic EM [38] and Monte Carlo EM (MCEM) [39] may be used to approximate the likelihood. In particular, incorporating EM with PF has long been proven to be effective in learning dynamical systems [40] and machine condition prognosis [41, 42].

The EM algorithm consists of iterating the expectation step (E-step) to create a function for the expectation of the log-likelihood evaluated using the current estimate, and the maximization step (M-step) to search for parameters maximizing the expected log-likelihood.

In the E-step, the density of the latent variables, $\pi(x_{0:n}|y_{0:n})$, needs to be evaluated, which in the case of state-space models corresponds to solving the smoothing problem.

In the following, we present how to use MCEM to estimate the unknown parameters for the artificial example in Section 2.3. The goal is to estimate θ_ϕ , θ_x and σ based on $y_{0:n}$ and $v_{0:n}$. To simplify the notations, we use x and y to denote $x_{0:n}$, $y_{0:n}$, respectively. Meanwhile, x_k and y_k still stands for the k th baseline and observed degradation level.

3.1. Initial guess

This section explains how to obtain the initial estimates, which serve as inputs to the EM, using the raw observation y and the TVOC information v . The parameters of the baseline degradation process, θ_x , and the parameters of observation density θ_ϕ and σ , are treated separately. Note that the method described here works only for the previously defined artificial example. For real data, as shown later in Section 4, other methods may exist depending on the dataset and understanding of the underlying physical system.

3.1.1. θ_ϕ

The degradation path formed by TVOC-dependent observations, $x + \Delta\phi$, is usually more fluctuating compared to the baseline degradation x . The changes in TVOC contribute to the ups and downs in the observations. Thus, a “good” estimation of θ_ϕ should make the inverse-transformed sequence, i.e., $y - \Delta\phi$, appear more monotonic than the original ones, y . Even if y is additionally perturbed by ϵ , we can search a θ_ϕ that maximizes the “monotonicity” of $y - \Delta\phi$. One possible measure of monotonicity is the rank correlation, e.g., Kendall’s τ or Spearman’s ρ , between time $t_{0:n}$ and $y - \Delta\phi$. Let $\theta_\phi^{(0)}$ be the initial guess of θ_ϕ , then,

$$\theta_\phi^{(0)} = \arg \max_{\theta_\phi} \tau(t, y - \Delta\phi|\theta_\phi) \quad (12)$$

3.1.2. θ_x

Having obtained $\theta_\phi^{(0)}$, we can have a rough estimation of the hidden degradation process: $\tilde{x} = y - \Delta\phi|\theta_\phi^{(0)}$. The initial guess of the parameters of Gamma process can be obtained using the isotonic regression, which fits a free-form line $\tilde{x}'$ to the sequence $\tilde{x}$. The fitted line, $\tilde{x}'$, is non-decreasing everywhere and lies as close to $\tilde{x}$ as possible. The regression problem can be solved using the pooled-adjacent-violators algorithm (PAVA) [43]. $\theta_x^{(0)}$ can then be obtained by maximizing the likelihood function associated with $\tilde{x}'$. However, as $\tilde{x}'$ is non-decreasing and may remain constant over certain intervals, it is necessary to remove the repeated values and the corresponding observation times. Most commercial software or packages have built-in functions to implement isotonic regression.

3.1.3. σ

The unobserved vector of measurement noises can be estimated as:

$$\tilde{\epsilon} = y - \tilde{x}' - \Delta\phi|\theta_\phi^{(0)} \quad (13)$$

The k th noise, $\tilde{\epsilon}_k$, is supposed to follow a normal distribution $\mathcal{N}(0, \sigma\|v_k\|_2)$. The initial guess for σ can be obtained by maximizing the likelihood:

$$\sigma^{(0)} = \arg \max_{\sigma} \sum_{k=0}^n -\log(\sigma\|v_k\|_2) - \frac{\tilde{\epsilon}_k^2}{2\sigma^2\|v_k\|_2^2} = \sqrt{\frac{1}{n+1} \sum_{k=0}^n \frac{\tilde{\epsilon}_k^2}{\|v_k\|_2^2}} \quad (14)$$

3.2. Monte Carlo EM

The estimation procedure consists in carrying out the following two steps iteratively: (1) given the current estimates θ_x , θ_ϕ , and σ , use a PF to obtain samples of the hidden baseline degradation path; and (2) update the estimates using these samples. The procedure terminates when convergence is reached. Details can be found in Algorithm 2.

Algorithm 2 Iterative parameter estimation

- 1: Obtain initial guesses $\theta_\phi^{(0)}$, $\sigma^{(0)}$, $\theta_x^{(0)}$;
- 2: **for** $m = 1 \dots M$ **do**
- 3: Use PF with parameter $\theta_\phi^{(m-1)}$, $\sigma^{(m-1)}$, $\theta_x^{(m-1)}$ to sample N paths: $x^{(i)}, i = 1 \dots N$
- 4: Compute $\theta_x^{(m)}$
- 5: Compute $\theta_\phi^{(m)}$ and $\sigma^{(m)}$
- 6: **end for**
- 7: The final estimators are the last elements of the estimate sequence.

3.2.1. Compute $\theta_x^{(m)}$ from the sampled paths

In step 4, $\theta_x^{(m)}$ is obtained by maximizing the likelihood function associated with the sampled paths $x^{(i)}, i = 1, 2, \dots, N$, computed from step 3. Note that these paths should be resampled so that each of them has the same weight. Let $\Delta x_k^{(i)} = x_k^{(i)} - x_{k-1}^{(i)}$ be the k th increment of the i th path during t_{k-1} and t_k . α_k can be defined following Eq. (6). The log-likelihood function of θ_x given $x^{(1)}, x^{(2)}, \dots, x^{(N)}$ is defined in Eq. (15):

$$L(\theta_x|x^{(1)}, \dots, x^{(N)}) = \sum_{i=1}^N \sum_{k=1}^n (\alpha_k - 1) \log(\Delta x_k^{(i)}) - \log(\Gamma(\alpha_k)) + \alpha_k \log(\beta) - \beta \Delta x_k^{(i)} \quad (15)$$

3.2.2. Update $\theta_\phi^{(m)}$ and $\sigma^{(m)}$

Given $x^{(i)}$ and y , we have, for $i = 1, 2, \dots, N$,

$$z_k^{(i)} = y_k - x_k^{(i)} = \Delta\phi_k|\theta_\phi + \epsilon_k \sim \mathcal{N}(\Delta\phi_k|\theta_\phi, \sigma\|v_k\|_2) \quad (16)$$

and the associated log-likelihood is:

$$L(\theta_\phi, \sigma|x^{(1 \dots N)}, y) = \sum_{i=1}^N \sum_{k=0}^n -\log(\sigma\|v_k\|_2) - \log(\sqrt{2\pi}) - \frac{(z_k^{(i)} - \Delta\phi_k|\theta_\phi)^2}{2\sigma^2\|v_k\|_2^2} \quad (17)$$

When the initial degradation value is a linear combination of the TVOC, i.e., $\phi(v_k|\theta_\phi) = v_k^T \theta_\phi$, the MLE estimates of θ_ϕ and σ can be explicitly derived. The derivatives of L with respect to $\theta_\phi[j]$ is given by:

$$\frac{\partial L(\theta_\phi, \theta_h|x^{(1,2,\dots,N)}, y)}{\partial \theta_\phi[j]} = \sum_i \sum_k \frac{z_k^{(i)} v_k[j]}{v_k^T v_k} - \sum_i \sum_k \frac{\theta_\phi^T v_k v_k[j]}{v_k^T v_k} \quad (18)$$

Let V be a $p \times (n+1)$ matrix (as the observation starts at 0 and ends at n) with $V[j, k]$ representing the value of the j th TVOC at moment k . Thus $v_k = V[:, k]$ is the k th column of V . Define $\tilde{V}$ as a $p \times (n+1)$ matrix with

$$\tilde{V}[j, k] = \frac{V[j, k]}{\sum_{i=1}^p V[i, k]^2} = \frac{V[j, k]}{v_k^T v_k}$$

and let U be a column vector of length p with

$$U[j] = \frac{1}{N} \sum_i \sum_k \frac{z_k^{(i)} V[j, k]}{v_k^T v_k}$$

Letting the derivatives of L with respect to $\theta_\phi[j]$ be 0, we have the following linear equation system:

$$V \tilde{V}^T \theta_\phi = U \quad (19)$$

which gives the MLE estimate of θ_ϕ as $(V \tilde{V}^T)^{-1} U$. Then, straightforwardly, the MLE estimate of σ is given by:

$$\sigma^{(m)} = \sqrt{\frac{1}{N(n+1)} \sum_i \sum_k \frac{(z_k^{(i)} - v_k^T \theta_\phi)^2}{v_k^T v_k}} \quad (20)$$

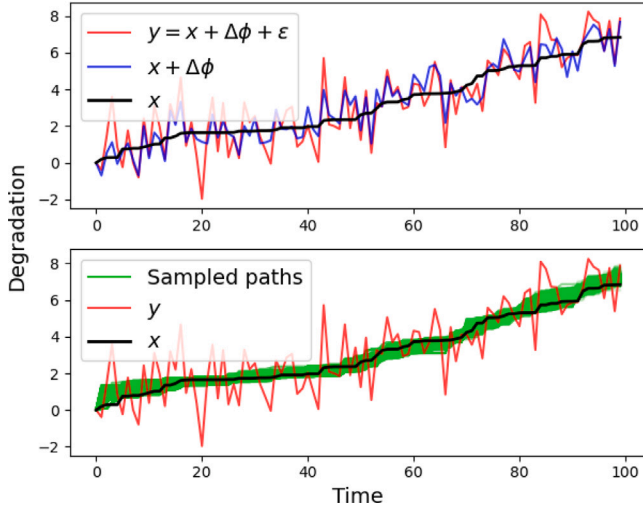


Fig. 2. Hidden and observed degradation paths are shown on the top. Baseline degradation x is drawn in black. The observed signal is perturbed by an observation bias related to TVOC (blue) and a noise vector ϵ (red). Sampled paths $z^{(i)}$ obtained from the PF are drawn in green.

3.3. A numerical example

Consider now the following configuration: $\theta_x = [\alpha, \beta, \eta] = [1, 4, 2]$, $\theta_\phi = [-2, 2]$, $\sigma = 1$, $n = 99$, and the total number of observations is 100. $t_i = i, i = 0, 1, \dots, n$ are the observation times. The parameter α and β are the baseline parameters in the gamma process, and η determines how the TVOC influences the shape parameter of the distribution of the Gamma process's increments, as in Eq. (6). The operating conditions v are sampled from a two-dimension uniform distribution $U[0, 1]$. In Fig. 2, top subfigure, the baseline degradation path x is drawn in black which is monotonically increasing. The measurements are first influenced by the operating conditions (blue curve) then contaminated by Gaussian noise (red curve y).

Using the methods in Section 3.1, the initial guess for the parameters are obtained as $\theta_x^{(0)} = [0.08, 1.04, 0]$, $\theta_\phi^{(0)} = [-1.86, 2.32]$, and $\sigma^{(0)} = 1.13$. In the bottom right subfigure of 3, the rank correlation, Kendall's τ , is plotted as a function of θ_ϕ . The initial guess $\theta_\phi^{(0)}$ maximizes τ as in Eq. (12). Note that the isotonic regression cannot estimate η , so the initial guess for $\eta^{(0)}$ is simply 0.

Then, the MCEM is carried out to infer the parameters, as described in Algorithm 2. We set the particle number $N = 3000$, and the total number of iteration $m = 150$. The convergence of the estimates of σ , θ_ϕ and θ_x towards the true values are shown in Figs. 3, bottom left, top right and top left, respectively.

Using the PF and the estimated parameters, we can sample x from y . In Fig. 2, the sampled paths, $z^{(i)}, i = 1, 2, \dots, N$ are drawn in green and are situated quite close to the black curve, which is the hidden baseline degradation. This shows that the PF method, combined with EM, can effectively estimate the unknown parameters and the hidden degradation path from the noisy observations.

3.4. Accuracy of the estimates

Following the previous numerical example, we discuss the influence of noise magnitude and data length on the accuracy of the estimates. Four configurations are considered, namely $[\sigma = 0.05, n = 49]$, $[\sigma = 0.5, n = 49]$, $[\sigma = 0.05, n = 99]$, $[\sigma = 0.5, n = 99]$. The values of θ_x and θ_ϕ remain the same. For each configuration, the parameter estimation as in Section 3.3 is repeated 200 times. Since the triad of θ_x is intertwined in the estimation, it only makes sense to consider its joint distribution. For illustration purposes, we show how the bias and variance of estimation

of σ and $\theta_\phi[0]$ change according to different noise levels and data lengths.

Since the true value of $\theta_\phi[0]$ is -2 for all four configurations, estimation accuracy is reflected through the mean squared error (MSE): 0.002, 0.044, 0.004, 0.026. Apparently, large noise ($\sigma = 0.5$) leads to large MSE (0.044 and 0.026, compared to 0.002 and 0.004) through the variance of the estimates (the red and orange curves are much more flattened in Fig. 4). The benefits of increasing data length to improve accuracy are significant only when the noise level is high (reduced to 60% when $\sigma = 0.5$). Similar results can be observed for the estimation of σ . Note that, the credibility of these conclusions should be further verified by increasing the number of repetitions.

4. Case studies on choke valve erosion data

Consider the Cv deviation of a production choke of type Cv428 with a CC50FNR linear Plug & Cage Trim, i.e., the maximal attainable Cv is 428 at the 100% opening. According to DNV's recommended practice [4] Appendix D.2, the level of damage may be monitored by monitoring deviation (increase) in Cv. The valve is equipped with a Stepping Linear Choke Actuator (SLCA), supplying a pressure of 5000 psi. This valve is installed in one of the North Sea Oil fields, and the Cv data have been monitored since June 2020. The theoretical Cv is provided in the vendor file and serves as a baseline value. The observation covers 413 days, with 288 data in total.

In Fig. 5, top left, we present a time series of valve opening over a 413-day period. As this is the only TVOC used in this study, it is denoted v without bold font. The top right graph shows the original Cv data, where the blue curve represents the theoretical Cv and the bullet points correspond to recorded Cv values measured at different dates. The shallow-colored data points correspond to earlier recordings while the dark-colored data points correspond to more recent ones. Notably, outliers can be observed where the recorded Cv values deviate significantly from the theoretical Cv.

Following a consultation with Equinor's expert, we decided to exclude all data collected during day 170 and 300 from the analysis. This decision was motivated by the observation of a large magnitude of fluctuation in the Cv deviation during this period, as indicated in the bottom left sub-figure represented in light red. Further interpretation of the excluded data may require additional information such as actuator torque or maintenance logs.

After this exclusion, 189 data points remained, as shown in the bottom right figure. The dark curve in Fig. 5, bottom left, represents the Cv deviation y plot for the remaining data.

Another way to visualize the evolution of the Cv deviation is to use a colored quiver plot. In Fig. 6, the data points $(v_k, y_k), k = 0, 1, 2, \dots, n$ with $n = 188$ are linked by colored arrows. The color represents the measurement time: the deeper the color, the earlier the data is recorded. Fig. 6 highlights the observation bias due to changes in the operating condition: often, the rise of the Cv deviation is not due to the valve erosion but simply because the valve is switched to a smaller opening.

4.1. Estimation of the initial degradation level

From Fig. 6, one can intuitively assume that the initial degradation level, ϕ , has a straight-line or parabola shape and moves up (and possibly down due to noise) as a whole. In the following, we assume that ϕ is a parabola with $\phi(v|\theta_\phi) = k_1 v + k_2 v^2$ and show how to obtain an initial guess of its parameters efficiently.

Since ϕ is not a linear function of v , the grid search method presented in Section 3.1.1 may be inefficient, because it can be difficult to determine the range of the parameters. We introduce a simple regression method based on least squares. The idea is to consider all the increments in the inverse-transformed sequence, $y_k - \Delta\phi_k, k = 1, 2, \dots, n$, as noise. Let $\delta_k = y_k - \Delta\phi_k - y_{k-1} + \Delta\phi_{k-1}$. Treating δ_k as noise is

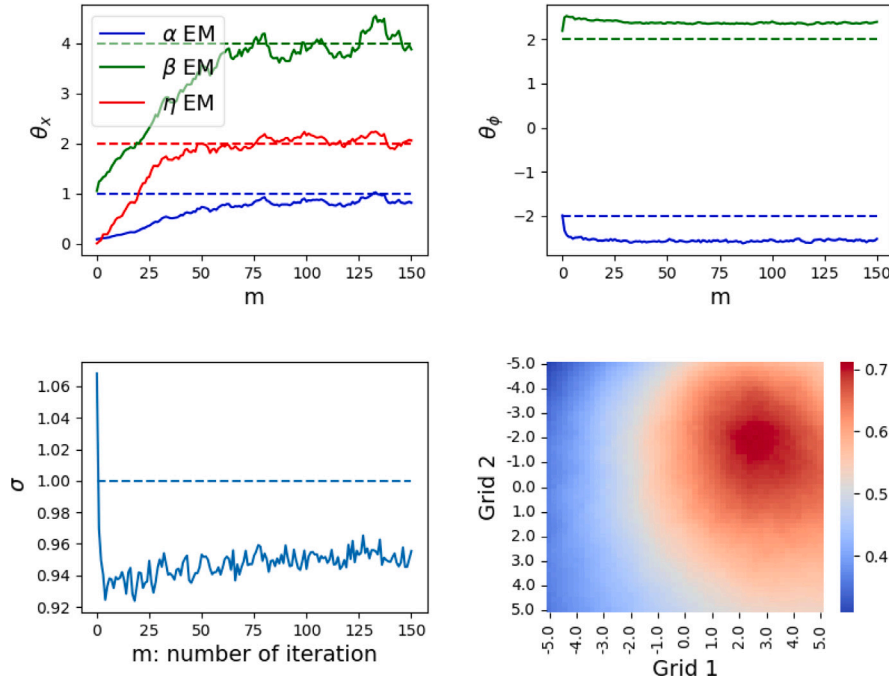


Fig. 3. Estimates of θ_x (top left), θ_ϕ (top right) and σ (bottom left). Convergence to the true value can be observed around 75 iterations. Kendall's τ as a function of θ_ϕ is shown in bottom right. A grid search is used to find θ_ϕ that maximizes τ (hottest point in the heatmap), which is then used as an initial guess.

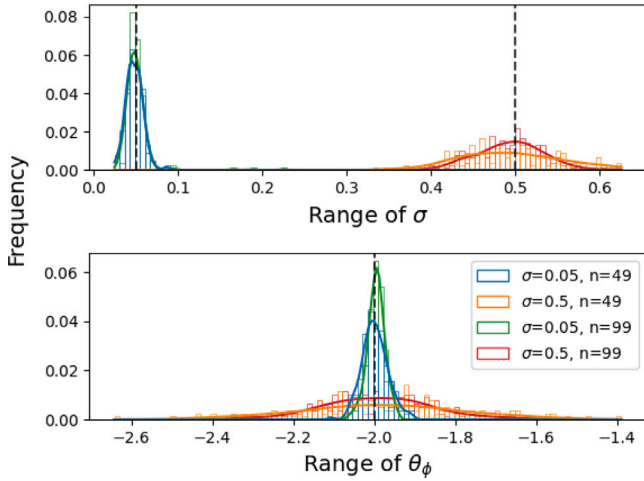


Fig. 4. Accuracy (bias and variance) of the estimates shown by their distribution. The top and bottom subfigures represent the distribution of estimates of σ and $\theta_\phi[0]$. Vertical lines show the true values: 0.05 and 0.5 for σ , and -2 for $\theta_\phi[0]$. The four configurations are shown in blue, orange, green and red, respectively.

apparently not accurate, since according to Eq. (1), $\delta_k = \Delta x_k + \Delta \epsilon_k$, which is the increment of the underlying baseline degradation process plus the difference between adjacent measurement errors. However, as shown in the following, minimizing the squared sum of δ_k can lead to a decent estimation of θ_ϕ .

Since ϕ is an order-2 polynomial, the closed form solution exists. Let U be a $n \times 2$ matrix with $U(i, j) = v_i^j - v_{i-1}^j$. $\delta = [\delta_1, \delta_2, \dots, \delta_n]^T = \Delta y - U\theta_\phi$ is the inverse transformed sequence. Then,

$$\theta_\phi^{(0)} = \arg \min_{\theta_\phi} \delta^T \delta = (U^T U)^{-1} U^T \Delta y \quad (21)$$

Note that ϕ can be defined as a higher-order polynomial of v , and Eq. (21) still works. It suffices to increase the number of columns of the matrix U . However, fitting a high-order polynomial could lead

to overfitting, which has been discussed in [44]. The least-squares estimates here is $\theta_\phi^{(0)} = [820, -2473]$. The intercept is not determined, and if we let the parabola cross the first observation point, which is (0.24, 5.39), then the initial degradation level is shown in Fig. 6 by the red dashed line. Notice that, at the beginning of the observation period, the valve is not good-as-new, and the initial degradation level is not 0.

4.2. Fitting a state-space model

The valve opening ranges from 0.19 to 0.33 during the observation period, so we choose 0.19 as the baseline TVOC. Once $\theta_\phi^{(0)}$ is calculated, we can transform all measurements to the baseline TVOC and obtain a rough estimation of the baseline degradation, $\bar{x}$, as shown by the black solid curve in Fig. 7. It is then rendered monotonic (red dash-dotted line, denoted as $\bar{x}'$) by the isotonic regression described in Section 3.1.2.

A generally increasing trend of the Cv deviation is shown in Fig. 6. Following existing research on choke valve erosion [5,7,45], we assume that the baseline degradation process is governed by a Gamma distribution. For other applications with little existing literature, statistical methods such as hypothesis testing or goodness-of-fit are advocated to determine the suitable baseline degradation model, e.g., Wiener process. Due to the presence of missing data (189 measurements for 413 days), it can be risky to assume that the opening remains the same when they are not recorded, so we assume the Gamma distribution parameters are independent of the valve opening. A more comprehensive model as in Eq. (6) should be considered if the valve opening is known for each day or at least for most days since we can complete the missing data using statistical tools such as the Last Observation Carried Forward method. The Cv calculation method ‘‘Sachdeva’’ takes into account the multiphase flow rate and other parameters, ensuring a reliable raw Cv (see Section 4.4.2 for more details). Therefore, the noise variance is considered constant, independent of the valve opening. Then, Eq. (5) and (7) becomes

$$\begin{cases} f(x|\theta, v_{k-1}) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} \exp(-\beta x) \\ \sigma_k = \sigma \end{cases} \quad (22)$$

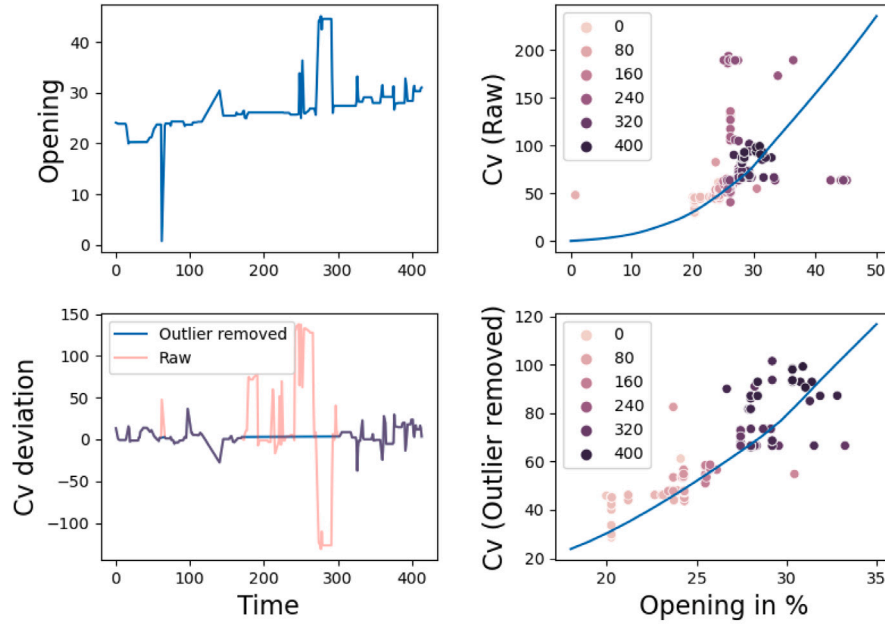


Fig. 5. Valve opening and Cv data. Top left: valve opening as a time series. Top right: the original Cv data with theoretical and recorded values. Outliers are visible in the recorded data. Bottom left: Cv deviation over time. The light red portion indicates the data that were excluded from the analysis. Bottom right: the remaining cleaned data after exclusion of the outliers.

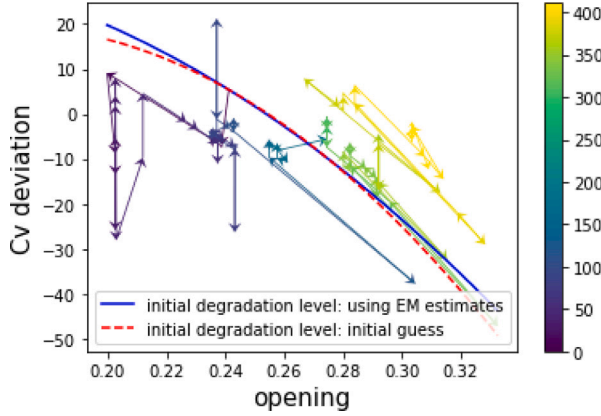


Fig. 6. The evolution of Cv deviation curves as a function of time is visualized by colored vectors: observations gathered at the early phase are linked by dark arrows, whereas those gathered near the end of observation are connected by light yellow arrows. The red dashed line represents the initial guess of the initial degradation level ϕ , and the blue solid line is the initial degradation level computed using the EM estimates.

Thus, the model is governed by parameters $\theta_x = [\alpha, \beta]$, θ_ϕ and σ . Fitting a Gamma process to $\bar{x}'$, we can obtain the initial guess of gamma parameters: $\theta_x^{(0)} = [0.059, 0.592]$. The initial guess for σ is $\sigma^{(0)} = 7.64$. Next, the MCEM as described in Algorithm 2 is used to estimate the parameters. Setting number of particles to 5000 and number of iteration to 50, the final estimates are obtained as $\hat{\theta}_x = [0.07, 0.74]$, $\hat{\theta}_\phi = [282, -1421]$ and $\hat{\sigma} = 5.37$. The sampled paths $z^{(i)}, i = 1, 2, \dots, N$, estimated by the PF using these parameters are shown in Fig. 7 by the green lines. Since $\hat{\theta}_\phi$ is quite different from the initial guess $\theta_\phi^{(0)}$, we plot the initial Cv deviation using $\hat{\theta}_\phi$ in the blue solid line in Fig. 6. Compared to the red dashed line, the blue one has a slightly smaller curvature, but the overall shape has not changed much.

If a failure threshold D is provided for the Cv deviation at the baseline TVOC, $z^{(i)}$ can be used to obtain the distribution of the RUL. Let $z_n^{(i)}$ be the last element in the i th sampled path. The survival

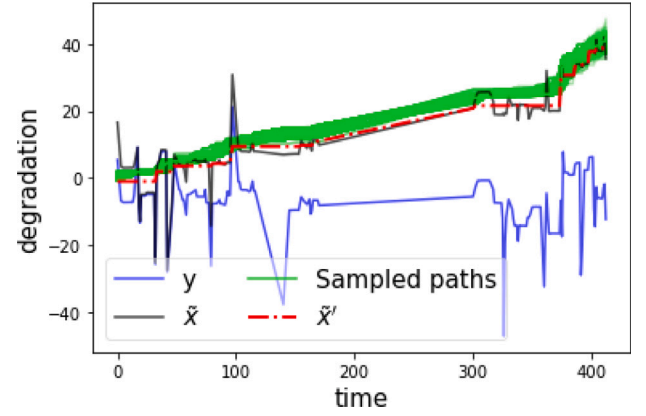


Fig. 7. Raw observation y is drawn in blue. $\bar{x}$, the initial guess of the baseline degradation based on least-squares estimates of $\theta_\phi^{(0)}$ is drawn in black. The red dashed line shows $\bar{x}'$ which is the output of isotonic regression. Green lines $z^{(i)}$ are the sampled paths computed using EM estimates.

function of RUL is therefore approximated by:

$$R_{\text{RUL}}(h) = \frac{1}{N} \sum_{i=1}^N \frac{\Gamma(\alpha h, \beta(D - z_n^{(i)}))}{\Gamma(\alpha h)} \mathbb{I}(z_n^{(i)} < D). \quad (23)$$

where $\Gamma(s, x)$ is the lower incomplete Gamma function: $\Gamma(s, x) = \int_0^x u^{s-1} e^{-u} du$.

4.3. Comparison with other methods

In previous studies on the erosion monitoring of choke valves, the opening is usually overlooked, and the non-monotonicity is ascribed to measurement noise [7]. Fitting a Gamma process to the raw data (blue curve in Fig. 5) is infeasible since a Gamma process can only depict monotone degradation. Wiener or noisy Gamma processes are potential candidates to describe the raw data. In a Wiener process with drift μ , the degradation is given by:

$$X_t = \mu \Lambda(t) + \sigma W(\Lambda(t)), \quad \Lambda(t) = t^\gamma, \quad (24)$$

where $\Lambda(t)$ is referred to as *time transformation* in [46] and W stands for the Brownian motion. $\Lambda(t)$ controls the shape of the degradation. Direct MLE can be used to estimate model parameters: $\hat{\mu} = -0.038$, $\hat{\gamma} = 1.02$, $\hat{\sigma} = 6.99$, which shows not only a decreasing drift, but also a significant variance. From a pure forecasting point of view, the decreasing trend of the raw Cv deviation can indeed happen if the valve opening continues to increase. Nevertheless, the Wiener model fails to reveal the relation between the operating condition and the health indicator, and gives misleading results about the health state of the choke valve. The other candidate model, noisy Gamma process, is inapplicable to this particular dataset: as the last observation is smaller than the first, the “hidden” degradation, if modeled by a Gamma process, would be a horizontal line that shows no trend at all.

Consider now the third candidate model introduced in [33] to investigate the RUL prediction for wind turbines under continuously varying rotating speeds. In both their problem and ours, there is only one single continuous TVOC. However, their method relies on the frequent reappearance of a baseline condition over the observation period. Otherwise, it would be difficult to transform the raw observations into the baseline speed, as described in their Algorithm 1. For wind turbines that have a periodic speed variation, this does not cause any problems. For subsea choke valves, however, the opening is often not recurrent, and does not necessarily return to exactly the same value, even if it varies in an interval.

4.4. Discussion on the choke valve condition monitoring

4.4.1. Slugging

From the data presented, it is highly likely that the well in question is experiencing slugging. This is evident as the well is constantly restricted at very low choke openings. Typically, the operation wants to increase the back pressure of the well and try to eliminate the slug flow regime when this occurs. However, without reviewing the time series plots, it is difficult to determine why the operation was restricting the choke valve. It is important to note that when the valve is partially closed, production is done under suboptimal conditions, and only safety aspects justify it. One way to monitor the slugging is to use pressure drop: if the flow is slugging, the pressure drop will be significantly affected by accumulated gas or liquid that arrives at different times at the valve.

Furthermore, slugging will potentially lead to issues in valve operation: the actuator does not respond correctly to the command of opening/closing the valve, resulting in the effective pass area in the valve not changing in accordance with the theoretical Cv curve. This may explain why sometimes the Cv is observed to be smaller than the theoretical value (Fig. 5). However, if such behavior is “permanent” due to valve geometry changes, then there could be “unbalanced” Cv growth [44], which makes the Cv deviate more at smaller openings than at larger openings.

4.4.2. Accuracy of the health indicator

We have used Cv deviation as the health indicator in this paper. Thus, Cv accuracy and theoretical Cv value are of vital importance. The Cv calculation method used here is called “Sachdeva”, and incorporates multiphase flow, gas-liquid ratio and the amount of produced water, which ensures that the calculated Cv is reliable. On the other hand, the theoretical Cv provided by manufacturers is typically estimated with fluids different from those encountered in an oil field. Manufacturers often use water to estimate the Cv of their equipment, making it challenging to compare the factory Cv with the installed Cv. This issue shall be addressed in our future study.

4.4.3. Failure threshold

According to Integrated Operations in the High North industrial report [47], when the Cv deviation finally passes a predefined threshold (typically, 7, which is also documented in [5]), inspections should be carried out. Such an inspection and replacement policy are actively used for Gullfaks and Statfjord fields in the Norwegian sector of the North Sea operated by Equinor. Note that The Cv deviation threshold should depend on the size and type of the choke valve. For example, for a small choke valve with maximal Cv = 15, a Cv deviation equaling 7 means that the choke is severely eroded. Moreover, in other industries, the failure threshold could be determined based on operating conditions.

4.4.4. Well test and model validation

Visual inspection is difficult for the subsea non-retrievable choke valve in our case study, but well tests can still reveal (to a more accurate extent) its Cv and therefore the damage level. In other applications, valves are often under partial/full stroke tests to detect leakage or other deterioration. Currently, we do not have test data for the investigated valves. But given further data, the validity of the model’s estimation of hidden degradation is verifiable.

4.4.5. Other influencing factors and production optimization

Apart from percent travel (opening), there are four major categories of factors that may influence the valve erosion: operating conditions (high temperature, high pressure), erosive agent (sand, barite, proppants), fluid properties (velocity, density, viscosity, and corrosiveness) and valve design and material. Meanwhile, in DNV RP O501, section 4.4 Erosion response model, the sand amount and speed are shown to be influencing factors. Not having the flow rate (which determines the sand speed) nor the sand amount is a weakness in the current case study. This highlights the importance of collecting and utilizing production data (e.g. velocity of multiphase flow) in degradation modeling.

The method described in this paper contributes to more accurate RUL estimation by bridging production (which is usually determined/influenced by TVOC) and degradation. Condition-based maintenance (CBM) then allows for joint optimization of maintenance and production, which implies slowing down the production to extend its lifetime. Although CBM is out of the scope of the current paper, we intend to address this issue in future work for not only choke valves but also general systems.

5. Conclusion

This paper presents a method that estimates the health state of a component working under continuous TVOC based on state-space models. Compared to most existing works where only one discrete operating condition is considered, our model accounts for multi-dimensional and continuous TVOC which influences not only the degradation increment but also observation bias and measurement error. The method is successfully applied to industrial choke valve condition monitoring data, demonstrating how to separate and quantify the Cv deviation growth due to erosion and valve opening changes. We also give a meaningful discussion on the condition monitoring for subsea choke valves, providing insights for practitioners in the O&G industry.

Below we show some potential future research directions. First, our model relies heavily on field experts’ opinions on the model structure. It is worth investigating statistical methods for model selection, such as hypothesis testing or goodness-of-fit analysis. These are particularly useful when no prior knowledge of model structure is available. Second, the baseline degradation is currently assumed to have a translation effect on the initial degradation level. Generalizing this assumption to include both scaling and translation as in [33] will lead to more flexible and realistic models that can be more widely applied. Third, adaptive

noise estimation with particle filter [48] could be a promising alternative method for noise modeling, which releases the assumption that the noise level is a deterministic function of the operating condition. Fourth, modeling the evolution of external environment variables with e.g., Markov models could be an important step towards an accurate RUL estimation, which is the base for joint optimization of production and maintenance.

In particular, for choke valves, we intend to extend the current model to include more process data, such as multiphase flow rates and well pressure to better understand the impact of production/operation on the Cv deviation. Physic-based or hybrid models can also be a promising alternative to the current model to incorporate phenomena such as slugging.

CRediT authorship contribution statement

Xingheng Liu: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Conceptualization. **Erling Lunde:** Investigation, Resources, Validation. **Fabio Diehl:** Investigation, Visualization. **Aibo Zhang:** Writing – review & editing, Writing – original draft, Conceptualization. **Jørn Vatn:** Supervision, Resources, Funding acquisition, Conceptualization. **Shen Yin:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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